



Fig. 15: dist installer window



Fig. 16: dist application

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

inverter can be obtained by dividing the total power by power factor as follows:

$$VA = \text{Total power} / \text{Power factor} = 550W / 0.7 = 785.7VA$$

But 785.7VA is not a standard rating for any inverter available in the market. So you should select an inverter having a rating of next higher available value, such as 800VA.

Calculating battery capacity. The next step is to decide which battery to buy. The type and capac-

ity of battery would depend on the back-up time you require. A domestic inverter normally uses a 12V battery, but the batteries have different ampere-hour (Ah) ratings. If you want an inverter battery that gives three-hour back-up time, you can calculate the battery capacity requirement as follows:

$$\text{Battery capacity} = (\text{Total power} \times \text{Back-up time}) / \text{Battery volts}$$

In this case,

$$\text{Battery capacity} = (550 \times 3) / 12 = 137.5Ah$$

Here Ah stands for ampere-hours.

But again, 137.5Ah is not a standard rating for the batteries available in the market. So, again, you should buy a standard battery with ratings slightly higher than this value, such as battery with 12V, 140Ah rating. The program output for this example is shown in Fig. 2.

Making a standalone application

You may not have Python software installed in your system to run this application. In that case, follow the steps given below to make a standalone application, so that you can run the program even without the software.

1. Download and install py2exe from the link <http://sourceforge.net/projects/py2exe/files/py2exe/0.6.9/>.

You can choose either 64-bit or 32-bit depending on the PC you have.

2. Create a setup.py file by downloading it from EFY DVD or from source.efymag.com. In the setup.py file, enter the script name as inverterApplication.py as shown in Fig. 3.

If you choose Python, wxpython and py2exe for 64-bit PC, you need to do some changes in setup.py file as shown in Fig. 4. Replace processor Architecture="x86" with processor Architecture="x64"

3. Copy the inverterApplication.py and setup.py files into a folder, say, inverterselection in your hard drive as shown in Fig. 5.

4. Now go to command panel and type as shown in Fig. 6 for 32-bit or as shown in Fig. 7 for 64-bit.

If there is 'msvcvp90.dll missing' error message, you can download the missing .dll file from <http://www.dll-files.com/dllindex/dll-files.shtml?msvcvp90> and put it in Python27/Dlls folder. If it still shows the same error, put the msvcvp90.dll file in c:/windows/SysWOW64 folder and reboot your system.

5. Now you should be able to see build and dist folders as shown in Fig. 8. If you open dist folder, you will see many other files, including inverterApplication file, as shown in Fig. 9.

6. Compress or zip the dist folder using WinRAR or a similar application as shown in Fig. 10.

The dist.zip file is created with reduced size as shown in Fig. 11.

7. To create an installer, you need to download NSIS software, which can be found at http://nsis.sourceforge.net/Main_Page. After installing the software, double click the NSIS icon and select 'Installer based on ZIP file' option as shown in Fig. 12.

Now, click Open button and browse your dist.zip file. Then click Generate button as shown in Fig. 13.

Finally, hit the Test button shown in Fig. 14.

After hitting the Test button the dist setup window will appear. Click Install button as shown in Fig. 15; dist application will be created as shown in Fig. 16.

Finally, delete both dist and build folders from the list shown in Fig. 16. Double click on dist application; an installation window will pop up. Click Install option. A new dist folder will be created. Under the dist folder you will find InverterApplication file. Double click on this file to run the application. You can also run the dist application file in some other compatible system without Python software. ●

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